Assignment 2

Due: March 8, 2025

1 Shreve Exercise 1.10

Let $\mathbb{P}$ be the uniform (Lebesgue) measure on $\Omega = [0, 1]$. Define

$$Z(\omega) = \begin{cases} 0 \text{ if } 0 \leq \omega < \dfrac{1}{2}, \\ 2 \text{ if } \dfrac{1}{2} \leq \omega \leq 1. \end{cases}$$

For $A \in \mathcal{B}[0, 1]$, define

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega).$$

(i) Show that $\widetilde{\mathbb{P}}$ is a probability measure.

(ii) Show that if $\mathbb{P}(A) = 0$, then $\widetilde{\mathbb{P}}(A) = 0$. We say that $\widetilde{\mathbb{P}}$ is *absolutely continuous* with respect to $\mathbb{P}$.

(iii) Show that there is a set A for which $\widetilde{\mathbb{P}}(A) = 0$, but $\mathbb{P}(A) > 0$. In other words, $\widetilde{\mathbb{P}}$ and $\mathbb{P}$ are not equivalent.

2 Shreve Exercise 1.14

(Change of measure for an exponential random variable) Let X be a nonnegative random variable defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with the *exponential distribution*, which is

$$\mathbb{P}\{X \leq a\} = 1 - e^{-\lambda a}, a \geq 0,$$

where λ is a positive constant. Let $\tilde{\lambda}$ be another positive constant, and define

$$Z = \frac{\tilde{\lambda}}{\lambda} e^{-(\tilde{\lambda} - \lambda)X}$$

. Define $\widetilde{\mathbb{P}}$ by

$$\widetilde{\mathbb{P}}(A) = \int_A Z d\mathbb{P} \text{ for all } A \in \mathcal{F}.$$

(i) Show that $\widetilde{\mathbb{P}}(\Omega) = 1$.

(ii) Compute the cumulative distribution function

$$\widetilde{\mathbb{P}}\{X \leq a\} \text{ for } a \geq 0$$

for the random vatiable X under the probability measure $\widetilde{\mathbb{P}}$.

3 Shreve Exercise 2.2

Independence of random variables can be affected by changes of measure. To illustrate this point, consider the space of two coin tosses $\Omega_2 = \{HH, HT, TH, TT\}$, and let stock prices be given by

$$S_0 = 4,\ S_1(H) = 8,\ S_1(T) = 2,$$
$$S_2(HH) = 16,\ S_2(HT) = S_2(TH) = 4,\ S_2(TT) = 1.$$

Consider two probability measures given by

$$\widetilde{\mathbb{P}}(HH) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(HT) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(TH) = \tfrac{1}{4},\ \widetilde{\mathbb{P}}(TT) = \tfrac{1}{4},$$
$$\mathbb{P}(HH) = \tfrac{4}{9},\ \mathbb{P}(HT) = \tfrac{2}{9},\ \mathbb{P}(TH) = \tfrac{2}{9},\ \mathbb{P}(TT) = \tfrac{1}{9}.$$

Define the random variable

$$X = \begin{cases} 1 & \text{if } S_2 = 4, \\ 0 & \text{if } S_2 \neq 4. \end{cases}$$

(i) List all the sets in $\sigma(X)$.

(ii) List all the sets in $\sigma(S_1)$.

(iii) Show that $\sigma(X)$ and $\sigma(S_1)$ are independent under the probability measure $\widetilde{\mathbb{P}}$.

(iv) Show that $\sigma(X)$ and $\sigma(S_1)$ are not independent under the probability measure $\mathbb{P}$.

(v) Under $\mathbb{P}$, we have $\mathbb{P}\{S_1 = 8\} = \frac{2}{3}$ and $\mathbb{P}\{S_1 = 2\} = \frac{1}{3}$. Explain intuitively why, if you are told that $X = 1$, you would want to revise your estimate of the distribution of S_1.

4 Shreve Exercise 4.8

(Solving the Vasicek equation) The Vasicek interest rate stochastic equation is

$$dR(t) = (\alpha - \beta R(t))dt + \sigma dW(t), \tag{1}$$

where α, β, and σ are positive constants. The solution to this equation is given in Example 4.4.10. This exercise shows how to derive this solution.

(i) Use (1) and the Itô-Doeblin formula to compute $d\left(e^{\beta t}R(t)\right)$. Simplify it so that you have a formula for $d\left(e^{\beta t}R(t)\right)$ that does not involve $R(t)$.

(ii) Integrate the equation you obtained in (i) and solve for $R(t)$ to obtain

$$R(t) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right) + \sigma e^{-\beta t}\int_0^t e^{\beta s}dW(s).$$